

Done — I added the new **CH2 / CH3 cooldown code** into the patch you pasted.

Download the updated full Arduino IDE patch here:

[PlantSong_TRUE_5VoicePoly_StableDelta_Cooldown.ino](#)

Added near the top:

```
const unsigned long CH2_COOLDOWN_MS = 2500;
const unsigned long CH3_COOLDOWN_MS = 6000;

unsigned long lastCh2Trigger = 0;
unsigned long lastCh3Trigger = 0;
```

Now Channel 2 and Channel 3 must pass **all three tests** before they play:

1. Raw delta must be high enough
2. Spike ratio must be high enough
3. Cooldown time must have passed

Current settings are:

```
const float CH2_DELTA_SPIKE_RATIO = 4.0f;
const float CH3_DELTA_SPIKE_RATIO = 9.0f;

const unsigned long CH2_MIN_RAW_DELTA = 180;
const unsigned long CH3_MIN_RAW_DELTA = 350;

const unsigned long CH2_COOLDOWN_MS = 2500;
const unsigned long CH3_COOLDOWN_MS = 6000;
```

For even less Channel 3, change:

```
const unsigned long CH3_COOLDOWN_MS = 9000;
```

That would allow Channel 3 only about once every 9 seconds maximum.